



MAVEN Departs

NASA's Mars Atmosphere and Volatile Evolution (MAVEN) satellite makes its way to Mars <http://dgit.in/flyMAVENfly>

Lazy Sun Sundrome?

The calm nature of the sun's surface may be something to be concerned about. Read on to know why <http://dgit.in/zzzsun>

Space Age

The pace of discovery really took a leap when Edward Hubble, from atop Mount Wilson, sighted a tiny nebula in the distant space outside the Milky Way. This led to the discovery of neighbouring galaxies, each with hundreds of billions of stars, and even more fertile possibilities for that one cosmic grain - the possibility of other planets like ours outside our solar system.

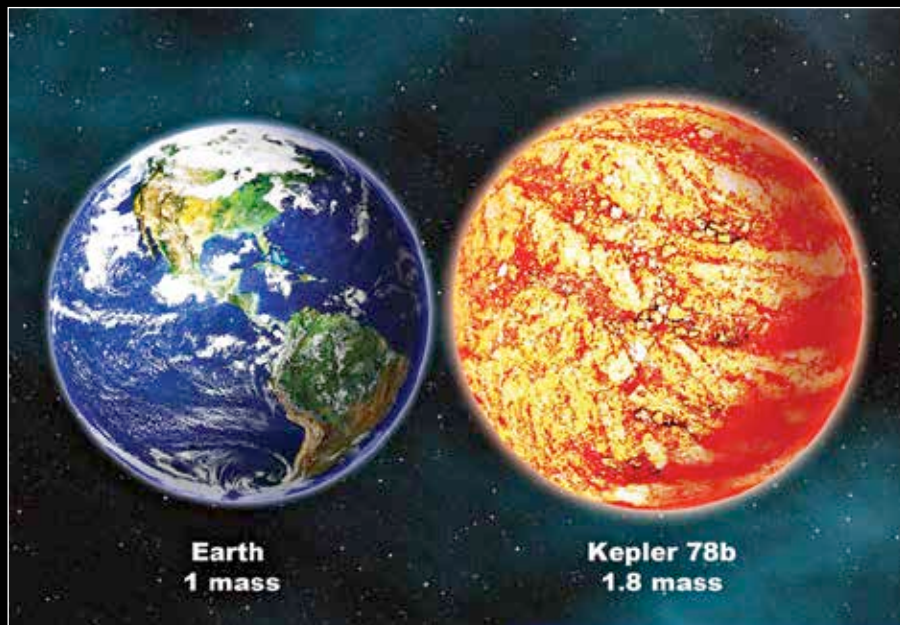
But it almost took another hundred years before a verifiable claim could be made over the discovery of an extrasolar planet or exoplanet. On a cold Pennsylvania morning in January 1994, a Polish astronomer by the name of Aleksander Wolszczan made the discovery that started off a new age of cosmic exploration - he discovered an exoplanet, two of them in fact. And the universe in all its whimsy made sure they were weird - these planet didn't orbit stars but pulsars - a dense rapidly spinning dead star - assuring that life wouldn't be possible there. But it was a start.

Since Wolszczan's discovery 1043 planets in 791 planetary systems have been discovered and verified, with thousands more awaiting review. These discoveries have been possible only due to the innovation and determination of an international assortment of scientists. And a very large USD \$550 million spacecraft known as Kepler.

The Kepler mission is quite literally the flagship in the search for Earth like planets. Already 4 years and 8 months in to its tumultuous 7-plus year voyage, the behemoth of a space observatory has already discovered thousands of possible Earth sized planets. And as the astronomers winnow through the pictures of the cosmos being sent back by this distant envoy they are already guesstimating that as many as 40 billion Earth-sized planets could be orbiting the habitable zones of stars within the Milky Way.

As is usually the case, when Gods want to punish us they grant us our wishes.

So in the long list of these planets a few discoveries deserve special mention. If not for their uniqueness, then for their evidence as proof that even rocket scientists don't know it all. Scientist have termed these discoveries as "weird" or "unusual" or "strange" or "anomalous" or simply "huh?".



Kepler78-b Earth's hellfire cousin the planet is only slightly larger by cosmic standards. The Kepler camera adds an extra 1.05356e25 pounds.

We take a look at five of the planets discovered so far that have the world's best minds reconsidering everything they learnt in Astronomy 101.

The Rebel Without A Cause: Kepler-78b

This fine specimen is the very definition of a planet that shouldn't exist. A scorching hell planet like Mustafar (2,830 °C; 5,120 °F peak surface temperature - Revenge of the Sith anyone?), Kepler-78b orbits its star (Kepler-78) at a distance 40 times closer than Mercury is to the sun. Remarkably similar to Earth in its composition, with 69% more mass and 20% greater size, Kepler-78b orbits causes



Show me your birth certificate Mr. HR8799b!

a "wobble" (yes, thats a technical term) in its star's orbit. Since its discovery in mid-2013 the planet's origin and evolution have remained a mystery for science can't fathom how it could've evolved in its current location or how it may have arrived there. Spooky.

The Uncomfortably Close Twins: Kepler-36c and Kepler-36b

These two planets defy all approximation of binary trajectories as they dance dangerously close to each other in their respective orbits around Kepler-36. Moving to an intimate rhythm, the two planets end up in perfect alignment every 97 days with less than 1.2 million miles between them, thats nearly 27 times less than the difference between the orbits of Earth and its closest neighbour Venus. The planets are also very dissimilar in their composition; one of them only 50% larger than Earth with a dry rocky make up while the other is a gaseous ball four times bigger than Earth. The never before seen proximity of these two planets in their orbit around its star had 46 scientists from Harvard and other less famous universities scratching their heads. They have yet to figure out how two heavenly bodies of such stark difference could end up in such close orbit to each other. Clearly they